Causal Inference for Error-Prone and Multi-Source Data

Dissertation Defense

Keith Barnatchez

Committee: Drs. Rachel Nethery, Giovanni Parmigiani and Kevin Josey

February 6th, 2025

Roadmap

Introduction

Chapter 1

Chapter 2

Chapter 3

Discussion

Use of Real-World Data on the Rise

Growing push for a greater role of real-world evidence in clinical decision making

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Real-World Evidence — What Is It and What Can It Tell Us?

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Figure 1: NEJM opinion piece on RWE

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Figure 2: Signing of 21st Century Cures Act

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Idea: Supplement clinical trials with real-world data sources

Multi-Source Data on the Rise

...as well as a push for clinical prediction models to be evaluated in terms of their generalizability: we want models that perform well **beyond** the context they were trained on

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Figure 4: OHDSI data consortium

Motivation

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- Data not originally collected for study purposes, e.g. electronic health record (**EHR**) data
  - Budget friendly + potentially more representative of general patient populations
- **Multi-source** data
  - Opportunity to improve the *generalizability* of study results + prediction models, as well as their precision

Contributions

Particularly when goal is to perform causal inference, these types of data present unique opportunities, but also **challenges**:

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- Non-study data, e.g. electronic health record (EHR) data
  - Key variables frequently subject to severe measurement error and missingness
- Multi-source data
  - **Heterogeneity** in patient populations across sources
  - Different sets of measured variables, different clinical definitions

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These challenges manifest as sources of bias that can invalidate naive statistical and causal inferences

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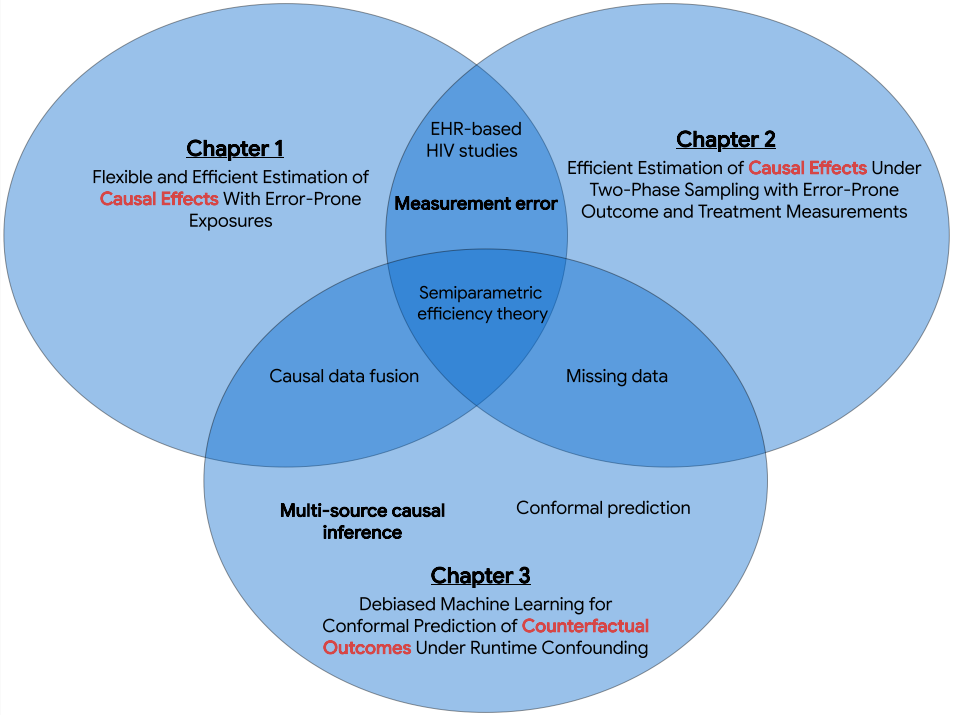
To facilitate the growing use of the types of data, this dissertation provides statistical tools for the estimation of causal effects in the presence of measurement error and multi-source data


Chapter 1
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Efficient Estimation of Causal Effects Under Two-Phase Sampling with Error-Prone Outcome and Treatment Measurements
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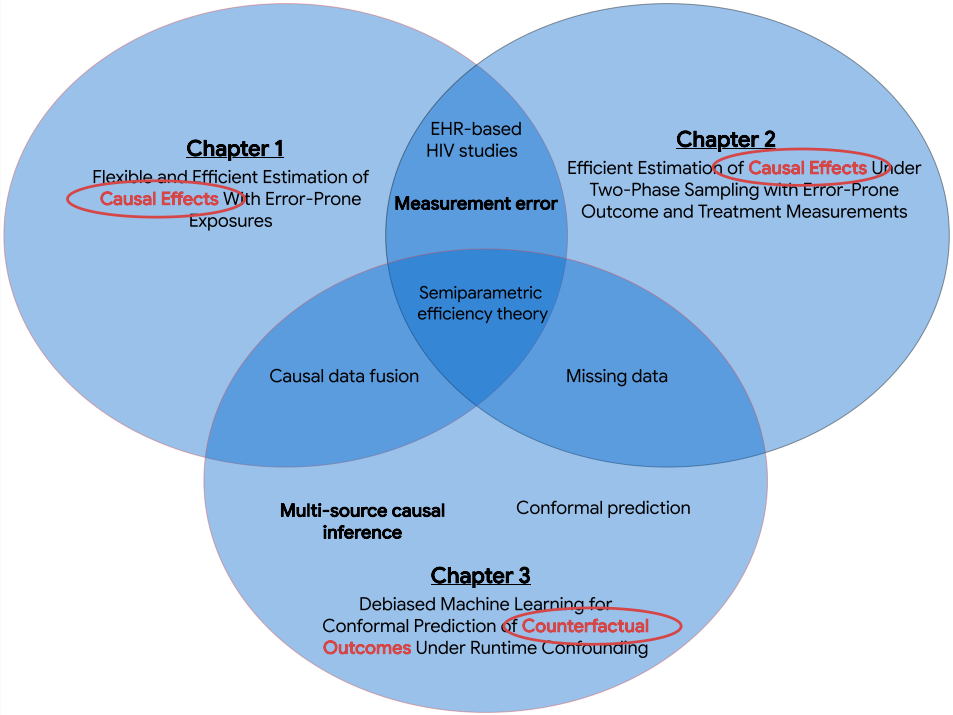
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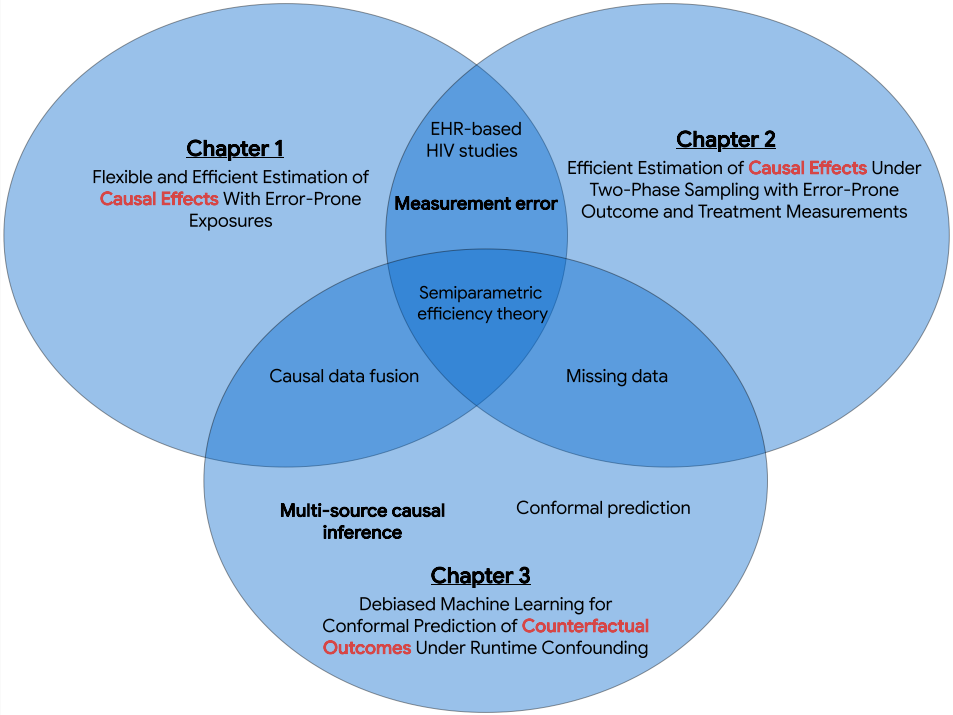
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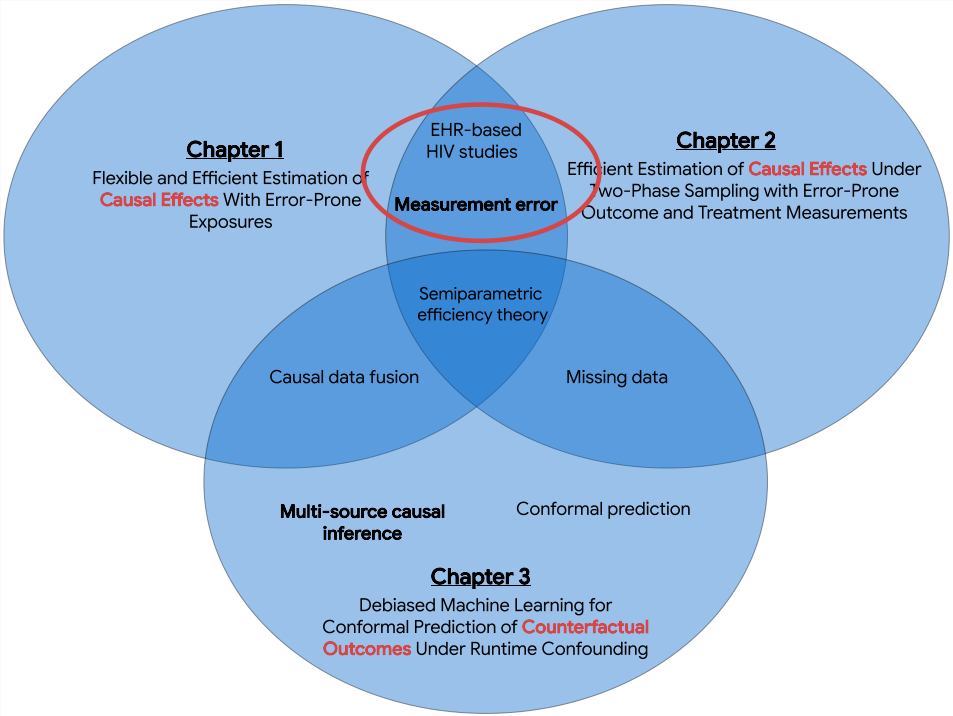
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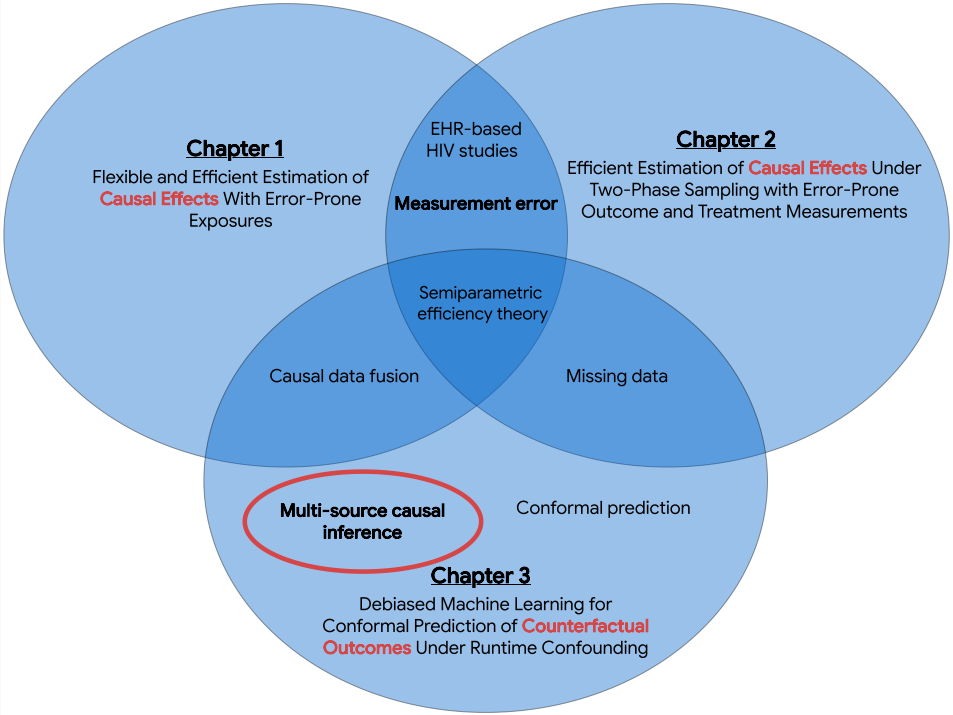
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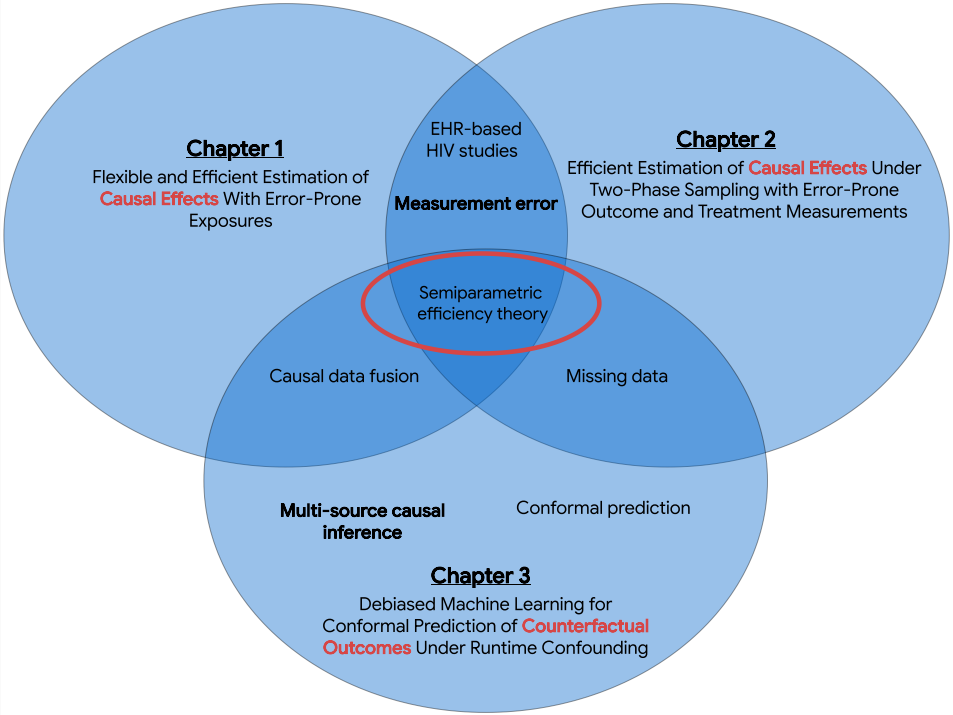
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Roadmap

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Flexible and Efficient Estimation of Causal Effects with Error-Prone Exposures: A Control Variates Approach for Measurement Error

Chapter 1 Go to Appendix

with Rachel C. Nethery, Bryan E. Shepherd, Giovanni Parmigiani and Kevin P. Josey

In press at *Biometrics*

Roadmap

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Problem Setting

The Control Variates Method

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Explosion of the use of electronic health record (EHR) data for conducting observational causal inference studies

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And for good reason!

- EHR data is typically cheaper to obtain, fairly representative of patient populations, rich in information on potential confounding factors $\boldsymbol{X}$, and big

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- EHR data is typically cheaper to obtain, fairly representative of patient populations, rich in information on potential confounding factors $\boldsymbol{X}$, and big

But EHR data tends to present numerous challenges

- Including measurement error in treatments/exposures (A) of interest

Motivation

In practice, EHR records often only contain error-prone measurements of A, denoted A^*, for every subject

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Y_1	?	A_1^*	$\boldsymbol{X}_1$
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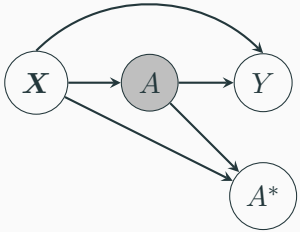

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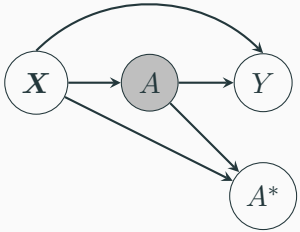

Growing literature on the perils of treatment measurement error in causal inference (Valeri 2021)

- Using A^* in place of A tends to produce biased effect estimates
- Difficult to correct for this bias without information on the measurement error mechanism
    - Requires *design-based* approaches that collect supplemental data

Addressing measurement error via study design

Y	A	A^*	$\boldsymbol{X}$	S
Y_1		A_1^*	$\boldsymbol{X}_1$	
Y_2		A_2^*	$\boldsymbol{X}_2$	
Y_3		A_3^*	$\boldsymbol{X}_3$	
Y_4		A_4^*	$\boldsymbol{X}_4$	
Y_5		A_5^*	$\boldsymbol{X}_5$	
Y_6		A_6^*	$\boldsymbol{X}_6$	

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Y	A	A^*	$\boldsymbol{X}$	S
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One design-based workaround: Spend additional time + resources to obtain gold-standard measurements of A for a **small subset** of the study data

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One design-based workaround: Spend additional time + resources to obtain gold-standard measurements of A for a **small subset** of the study data

- Typically referred to as a two-phase sampling study design (Hidiroglou 2001) or validation study
- An EHR contexts, can be performed through manual chart review

Addressing measurement error via study design

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The subset of data with gold-standard measurements is typically referred to as the *validation data*

- **Intuition**: provides complete information for a subset of data, and provides insight into measurement error mechanism

Contributions

In this paper, we present a flexible framework for constructing efficient estimators of causal effects in the presence of a partially-validated error-prone exposure/treatment variable

Our proposed framework is **user-friendly**, and enjoys numerous properties commonly associated with doubly-robust estimators

Roadmap

Chapter 1

- Motivation
- Problem Setting
- The Control Variates Method
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Exposure measurement error data structure

Causal estimand: $\tau \stackrel{\text{def}}{=} \mathbb{E}[Y(1) - Y(0)]$ (average treatment effect)

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(a) Main dataset

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(b) Validation dataset

Assumptions

Make the standard causal inference assumptions: **consistency**, **positivity** and **unconfoundedness**

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Additionally, assume the validation data is obtained completely at random: $S \perp\!\!\!\perp (Y, A, A^*, \boldsymbol{X})$

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- Can often be enforced by design in EHR data settings
- Can be relaxed to allow for more complex validation sampling schemes/study designs
  - Covariate-dependent: $S \perp\!\!\!\perp (Y, A, A^*) | \boldsymbol{X}$
  - Error-free variable dependent: $S \perp\!\!\!\perp A | (\boldsymbol{X}, A^*, Y)$

Estimation wishlist

Ideally, we'd like an estimator that

- Is **(1)** unbiased, **(2)** model-agnostic, and **(3)** efficient

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- But this approach completely ignores the remaining observations!
  - Unbiased, but highly inefficient as validation samples are typically small

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Our approach is adapted from Yang and Ding (2019)

- **Idea**: Improve the efficiency of an initial unbiased (but inefficient) estimator of τ by **augmenting** it with a variance reduction term formed from the full data
  - This variance reduction term is typically referred to as a control variate

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Step 1: obtain validation data only estimator

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Y_3	A_3	A_3^*	X_3
Y_5	A_5	A_5^*	X_5

Validation data only estimate: $\hat{\tau}_{\text{val}}$

Step 2: construct the control variate

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Error-prone estimate: $\hat{\tau}^{\text{e.p.}}_{\text{main}}$

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Error-prone estimate: $\hat{\tau}^{\text{e.p.}}_{\text{val}}$

Step 3: Compute the variance reduction term

Obtain $\hat{\Gamma} = \widehat{\text{Cov}}(\hat{\tau}_{\text{val}}, \hat{\tau}^{\text{e.p.}}_{\text{main}} - \hat{\tau}^{\text{e.p.}}_{\text{val}})$ and $\hat{V} = \widehat{\text{Var}}(\hat{\tau}^{\text{e.p.}}_{\text{main}} - \hat{\tau}^{\text{e.p.}}_{\text{val}})$

Step 4: Form the final estimator

Obtain final estimate: $\hat{\tau}_{\text{CV}} = \hat{\tau}_{\text{val}} - \hat{\Gamma}\hat{V}^{-1}(\hat{\tau}^{\text{e.p.}}_{\text{main}} - \hat{\tau}^{\text{e.p.}}_{\text{val}})$

Control variates method: properties

Efficiency gain: $\mathsf{Var}(\hat{\tau}_{\mathsf{CV}}) = \mathsf{Var}(\hat{\tau}_{\mathsf{val}}) - \Gamma^2/V$, where

- $\Gamma = \mathsf{Cov}(\hat{\tau}_{\mathsf{val}}, \hat{\tau}^{\mathsf{e.p.}}_{\mathsf{main}} - \hat{\tau}^{\mathsf{e.p.}}_{\mathsf{val}})$
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Double robustness: Consistent if either the outcome model or propensity score model is correctly specified

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Flexible estimation of nuisance models:

- $\hat{\tau}_{\text{CV}} \xrightarrow{p} \tau$ at $\sqrt{n}$ rates

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Flexible estimation of nuisance models:

- $\hat{\tau}_{\text{CV}} \xrightarrow{p} \tau$ at $\sqrt{n}$ rates
- Even if the nuisance models are estimated with ML methods that themselves have slower rates of convergence
  - Common property of "doubly-robust" estimators

Control variates method: extensions

The control variates method is **flexible** – can account for...

- More general validation data sampling schemes / account for multiple study sites
- Simultaneous error in the outcome of interest

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The Vanderbilt Comprehensive Care Clinic (VCCC)

EHR database with information on $\approx$ 1900 individuals living with HIV between 1998-2011 that began receiving care at the VCCC

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- Reliable measurements on demographic information, risk factors and baseline lab measurements (baseline covariates, $\boldsymbol{X}$)

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Causal estimand: Causal risk difference of suffering an AIDS-defining event at baseline (A) on 5-year (post initial visit) survival Y

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Goal: Compare the control variates estimator to (1) a validation data only estimator, (2) a naive AIPW estimator, and (3) an oracle AIPW estimator

VCCC application

"Reveal" increasingly larger shares of the validated data $\mathbb{P}(S = 1)$

- E.g. 5%, 10%, $15\%, \ldots$
- Implement proposed estimators at each share

Y	A	A^*	X	S
Y_1	A_1	A_1^*	X_1	1
Y_2		A_2^*	X_2	0
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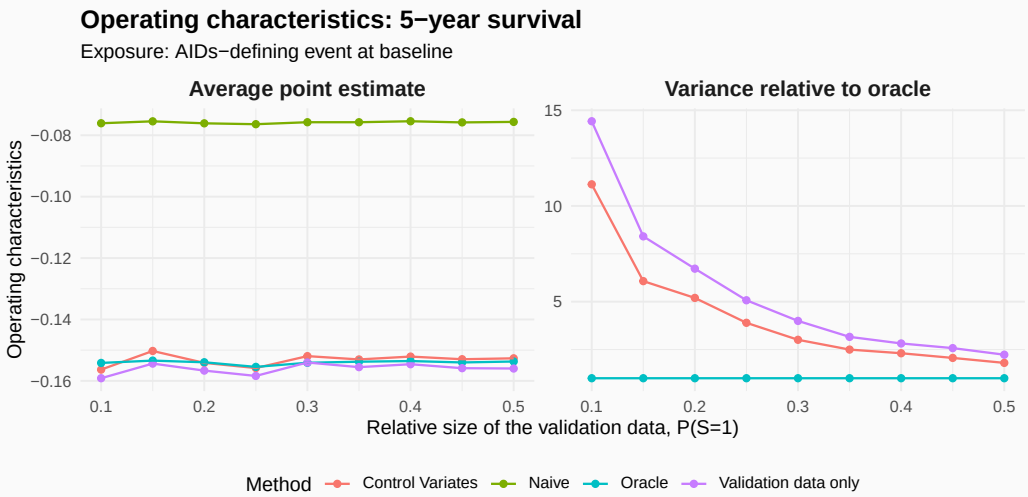
Results


Operating characteristics: 5−year survival
Exposure: AIDs−defining event at baseline
Average point estimate
Variance relative to oracle
Operating characteristics
−0.08
−0.10
−0.12
−0.14
−0.16
15
10
5
0.1
0.2
0.3
0.4
0.5
Relative size of the validation data, P(S=1)
Method
Control Variates
Naive
Oracle
Validation data only

Roadmap

Chapter 1

- Motivation
- Problem Setting
- The Control Variates Method
- Data Application
- Discussion

Discussion

- Growing set of methods for accounting for measurement error in causal inference

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The control variates method isn't appropriate for every measurement error problem

- Requires the availability of /ability to obtain a validation dataset
- As always, care should be taken to assess the plausibility of the causal assumptions

Roadmap

Introduction

Chapter 1

Chapter 2

Chapter 3

Discussion

Efficient Estimation of Causal Effects Under Two-Phase Sampling with Error-Prone Outcome and Treatment Measurements

Chapter 2 Go to Appendix

with Kevin P. Josey, Nima S. Hejazi, Bryan E. Shepherd, Giovanni Parmigiani and Rachel C. Nethery

Under review

Roadmap

Chapter 2

Background

Methods

Approach 1

Approach 2

Ensemble Estimator

Data Application

Discussion

Motivation

Chapter 1 provided a flexible framework for addressing treatment measurement error in the presence of a validation sample, using an EHR data application as a motivating example

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In practice, EHR-derived clinical outcomes are also commonly subject to measurement error

Further, validation data is increasingly collected according to complex **sampling mechanisms** that can complicate downstream statistical inferences

- Remains a need for the development of semiparametric efficient estimators of causal effects which accommodate complicated validation sampling mechanisms
- Particularly when both outcomes (Y) and treatments (A) are measured with error

EHR data: validating observations based on error-prone measurements

Y	Y^*	A	A^*	$\boldsymbol{X}$
NA	Y_1^*	NA	A_1^*	$\boldsymbol{X}_1$
NA	Y_2^*	NA	A_2^*	$\boldsymbol{X}_2$
NA	Y_3^*	NA	A_3^*	$\boldsymbol{X}_3$
NA	Y_4^*	NA	A_4^*	$\boldsymbol{X}_4$
NA	Y_5^*	NA	A_5^*	$\boldsymbol{X}_5$
NA	Y_6^*	NA	A_6^*	$\boldsymbol{X}_6$

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NA	Y_4^*	NA	A_4^*	$\boldsymbol{X}_4$
NA	Y_5^*	NA	A_5^*	$\boldsymbol{X}_5$
NA	Y_6^*	NA	A_6^*	$\boldsymbol{X}_6$

In practice, we often only have error-prone measurements of Y and A, denoted Y^* and A^*

- Well-documented that using Y^* and A^* in place of Y and A can lead to severely biased causal effect estimates

EHR data: validating observations based on error-prone measurements

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1

In practice, can sometimes spend time + money to obtain gold-standard measurements for a random (typically small) subset of this data

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Y_1	Y_1^*	A_1	A_1^*	X_1	1
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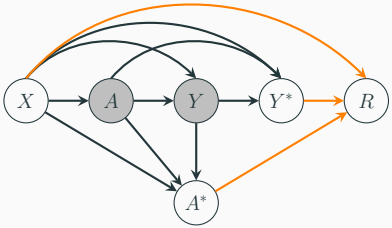

In practice, selection into this subset can often be controlled, and may depend on the initial error-prone measurements: $R \not\perp\!\!\!\perp A^*, Y^*, \boldsymbol{X}$

- Especially common sampling strategy if Y and/or A are rare

EHR data: validating observations based on error-prone measurements

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
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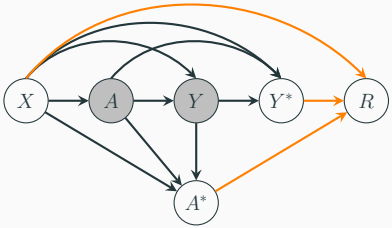

In practice, selection into this subset can often be controlled, and may depend on the initial error-prone measurements: $R \not\perp\!\!\!\perp A^*, Y^*, \boldsymbol{X}$

- **Intuition**: over-sample subjects who contribute more information to the target estimand

Our Contributions

At a high-level, our work addresses the following question: How do we estimate causal effects nonparametrically and efficiently when

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We present two asymptotically equivalent approaches to constructing efficient nonparametric causal effect estimators, leveraging their connections to construct estimators with improved finite-sample behavior

Roadmap

Chapter 2

Background

Methods

Approach 1

Approach 2

Ensemble Estimator

Data Application

Discussion

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Approach 1: Using the observed data distribution

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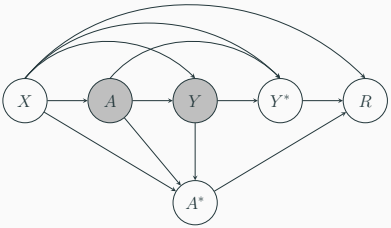

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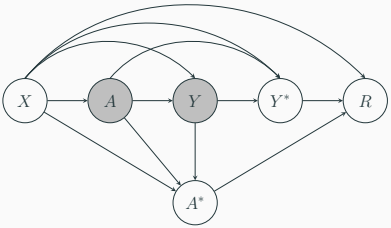

Main goal: Estimate counterfactual means $\mathbb{E}[Y(a)]$ $(a \in \{0,1\})$ *efficiently*, allowing R to depend on $\boldsymbol{X}, Y^*$ and A^*

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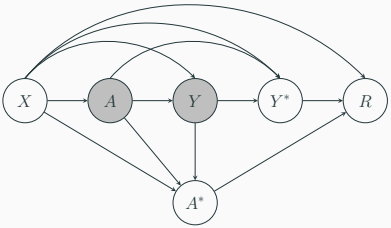

Construction of estimators follows standard semiparametric estimation pipeline:

causal estimand

$$\mathbb{E}[Y(a)] \xrightarrow[\text{Assumptions}]{} \psi_0 = \mathbb{E}[f(\text{data})] \xrightarrow[\text{stat. estimand}]{} \hat{\psi}_n^{\text{PI}} = \frac{1}{n}\sum_{i=1}^{n} f(\text{data}_i) \xrightarrow[\text{plug-in estimator}]{} \xrightarrow[\text{(tedious) work}]{} \xrightarrow[\text{debiased est.}]{} \hat{\psi}_n^{\text{OS}}$$

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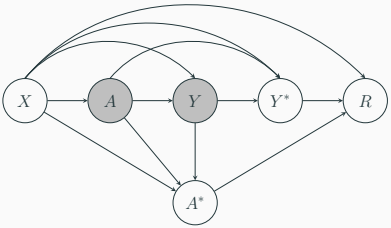

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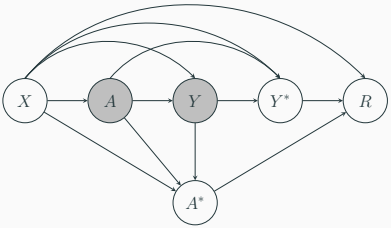

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stat. estimand

plug-in estimator

debiased est.

(tedious) work

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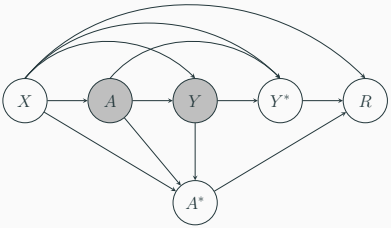

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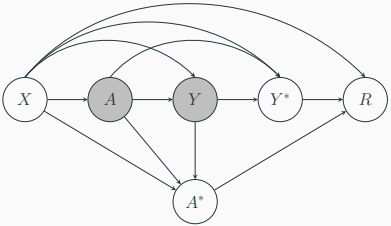

Letting $\boldsymbol{Z} = (\boldsymbol{X}, A^*, Y^*)$, we make the three "core" causal assumptions:

$$\textbf{(1)}\ \underbrace{Y = AY(1) + (1-A)Y(0)}_{\text{Consistency}}, \qquad \textbf{(2)}\ \underbrace{\mathbb{P}(A = 1|\boldsymbol{X}) \in (0,1)}_{\text{Positivity of treatment}}, \qquad \textbf{(3)}\ \underbrace{Y(a) \perp\!\!\!\perp A|\boldsymbol{X}}_{\text{Unconfoundedness}},$$

as well as two assumptions related to the missingness mechanism for Y^* and A^*

$$\textbf{(4)}\ \underbrace{\mathbb{P}(R = 1|\boldsymbol{Z}) \in (0,1)}_{\text{Validation positivity}}, \qquad \textbf{(5)}\ \underbrace{(Y^*, A^*) \perp\!\!\!\perp R|\boldsymbol{Z}}_{\text{Missing at random}}$$

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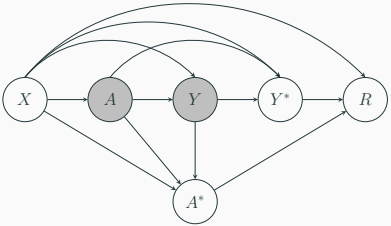

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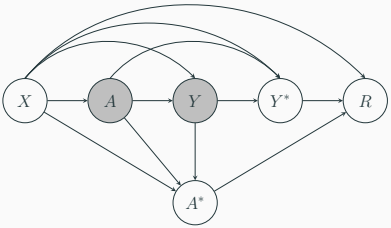

Letting $\boldsymbol{Z} = (\boldsymbol{X}, A^*, Y^*)$, we derive a plug-in estimator

Approach 1: Using the observed data distribution

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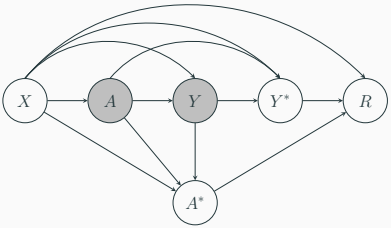
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



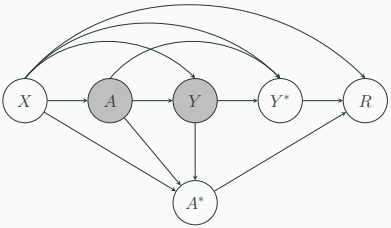
Letting $\boldsymbol{Z} = (\boldsymbol{X}, A^*, Y^*)$, we derive a plug-in estimator

$$\hat{\psi}_a^{\text{PI},1} = \frac{1}{n}\sum_{i=1}^{n} \frac{\hat{\mathbb{E}}[\hat{\lambda}_a(\boldsymbol{Z}_i) \cdot \hat{\mu}_a(\boldsymbol{Z}_i)|\boldsymbol{X}_i]}{\hat{\mathbb{E}}[\hat{\lambda}_a(\boldsymbol{Z}_i)|\boldsymbol{X}_i]}$$

where $\hat{\lambda}_a(\boldsymbol{Z}) = \hat{\mathbb{P}}(A = a|\boldsymbol{Z}, R = 1)$ and $\hat{\mu}_a(\boldsymbol{Z}) = \hat{\mathbb{E}}(Y|A = a, \boldsymbol{Z}, R = 1)$

Approach 1: Using the observed data distribution

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



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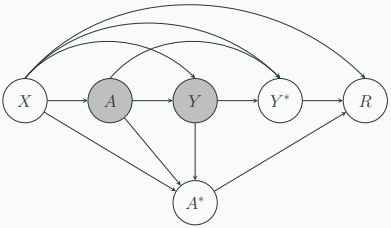
where $\hat{\lambda}_a(\boldsymbol{Z}) = \hat{\mathbb{P}}(A = a|\boldsymbol{Z}, R = 1)$ and $\hat{\mu}_a(\boldsymbol{Z}) = \hat{\mathbb{E}}(Y|A = a, \boldsymbol{Z}, R = 1)$

- Interpretation: IPW on imputed values, after marginalizing out post-treatment variables

Approach 1: Using the observed data distribution

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



Letting $\boldsymbol{Z} = (\boldsymbol{X}, A^*, Y^*)$, we derive a plug-in estimator

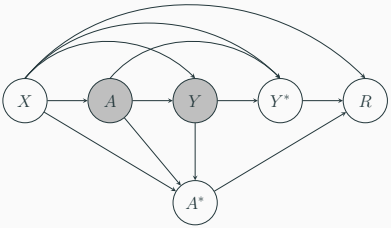
$$\hat{\psi}_a^{\text{PI},1} = \frac{1}{n} \sum_{i=1}^{n} \frac{\hat{\mathbb{E}}[\hat{\lambda}_a(\boldsymbol{Z}_i) \cdot \hat{\mu}_a(\boldsymbol{Z}_i) | \boldsymbol{X}_i]}{\hat{\mathbb{E}}[\hat{\lambda}_a(\boldsymbol{Z}_i) | \boldsymbol{X}_i]}$$

where $\hat{\lambda}_a(\boldsymbol{Z}) = \hat{\mathbb{P}}(A = a | \boldsymbol{Z}, R = 1)$ and $\hat{\mu}_a(\boldsymbol{Z}) = \hat{\mathbb{E}}(Y | A = a, \boldsymbol{Z}, R = 1)$

- Drawback: Inference intractable when nuisance models are fit data-adaptively

Approach 1: Using the observed data distribution

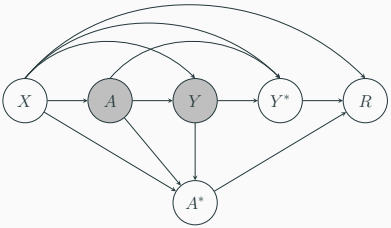
Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



To enable inference, we derive a one-step debiased estimator

Approach 1: Using the observed data distribution

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



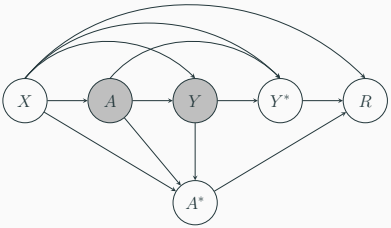
To enable inference, we derive a one-step debiased estimator

$$\hat{\psi}_a^{\text{OS},1} = \frac{1}{n}\sum_{i=1}^{n} \frac{\hat{\mathbb{E}}[\hat{\lambda}_a(\boldsymbol{Z}_i)\cdot\hat{\mu}_a(\boldsymbol{Z}_i)|\boldsymbol{X}_i]}{\hat{\mathbb{E}}[\hat{\lambda}_a(\boldsymbol{Z}_i)|\boldsymbol{X}_i]} + \widehat{\text{BC}}$$

where $\widehat{\text{BC}}$ is a bias correction term based on the efficient influence function for the statistical estimand, ψ_a

Approach 1: Using the observed data distribution

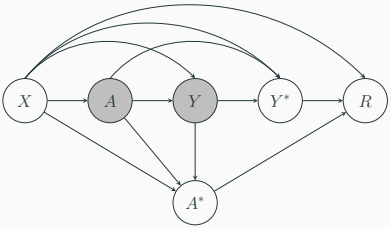
Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



We document multiple properties of $\hat{\psi}_a^{\text{OS},1}$

Approach 1: Using the observed data distribution

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1

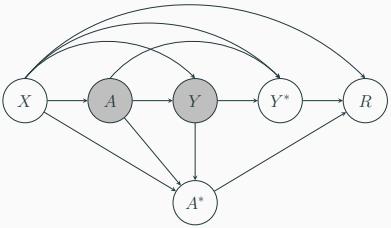


We document multiple properties of $\hat{\psi}_a^{\text{OS},1}$

1. Bias correction enables valid inference when nuisance models are fit with flexible ML methods that converge at $n^{1/4}$ rates, however...

Approach 1: Using the observed data distribution

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



We document multiple properties of $\hat{\psi}_a^{\text{OS},1}$

1. Bias correction enables valid inference when nuisance models are fit with flexible ML methods that converge at $n^{1/4}$ rates, however...

2. This bias correction term introduces numerous unstable weighting terms that can harm finite sample performance
   - Particularly concerning, as validation samples tend to be small in practice

Roadmap

Chapter 2

Background

Methods

Approach 1

Approach 2

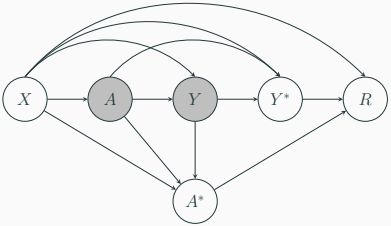
Ensemble Estimator

Data Application

Discussion

Approach 2: Complete-data projection

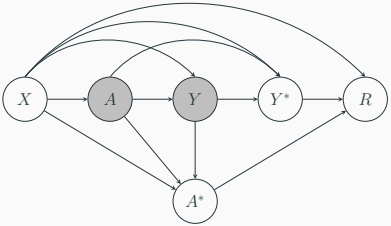
Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



Approach 2 is based on a well-developed, but relatively underutilized, framework for constructing debiased estimators under missing data (van der Laan and Robins 2003)

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
	Y_2^*		A_2^*	X_2	0
	Y_3^*		A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
	Y_5^*		A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1

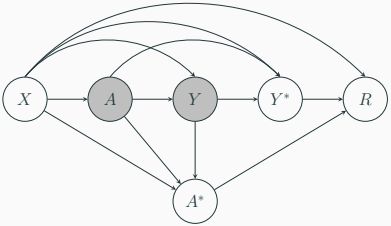


Approach 2 is based on a well-developed, but relatively underutilized, framework for constructing debiased estimators under missing data (van der Laan and Robins 2003)

- Links (i) the estimator we'd ideally construct under complete data to (ii) the observed data structure, where Y and A are partially missing

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
Y_3	Y_3^*	A_3	A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
Y_5	Y_5^*	A_5	A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



Idea: If we had complete data, could construct a plug-in estimator $\hat{\psi}_a^{\text{PI},2} = \frac{1}{n}\sum_{i=1}^n \hat{m}_a(\boldsymbol{X}_i)$,

$$\hat{\psi}_a^{\text{OS},2} = \hat{\psi}_a^{\text{PI},2} + \frac{1}{n}\sum_{i=1}^n \left(m_a(\boldsymbol{X}_i) + \frac{I(A_i = a)}{g_a(\boldsymbol{X})}\{Y_i - m_a(\boldsymbol{X}_i)\} - \hat{\psi}_a^{\text{PI},2} \right)$$

where $m_a(\boldsymbol{X}) = \mathbb{E}(Y|A = a, \boldsymbol{X})$ and $g_a(\boldsymbol{X}) = \mathbb{P}(A = a|\boldsymbol{X})$

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
Y_3	Y_3^*	A_3	A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
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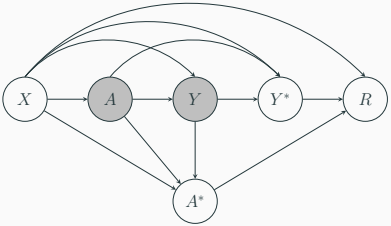

Idea: If we had complete data, could construct a plug-in estimator $\hat{\psi}_a^{\mathsf{PI},2} = \frac{1}{n}\sum_{i=1}^n \hat{m}_a(\boldsymbol{X}_i)$, as well as an augmented inverse probability weighted (AIPW) estimator

$$\hat{\psi}_a^{\mathsf{OS},2} = \hat{\psi}_a^{\mathsf{PI},2} + \frac{1}{n}\sum_{i=1}^n \left(\hat{m}_a(\boldsymbol{X}_i) + \frac{I(A_i = a)}{\hat{g}_a(\boldsymbol{X}_i)}\{Y_i - \hat{m}_a(\boldsymbol{X}_i)\} - \hat{\psi}_a^{\mathsf{PI},2} \right)$$

where $m_a(\boldsymbol{X}) = \mathbb{E}(Y|A = a, \boldsymbol{X})$ and $g_a(\boldsymbol{X}) = \mathbb{P}(A = a|\boldsymbol{X})$

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
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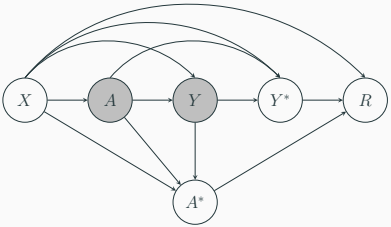
Y_6	Y_6^*	A_6	A_6^*	X_6	1



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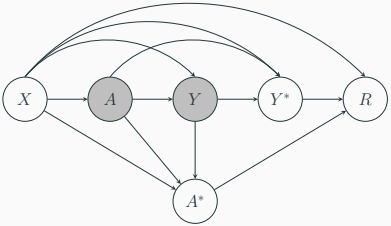
$$\hat{\psi}_a^{\mathsf{OS},2} = \hat{\psi}_a^{\mathsf{PI},2} + \frac{1}{n}\sum_{i=1}^n \left(\hat{m}_a(\boldsymbol{X}_i) + \frac{I(A_i = a)}{\hat{g}_a(\boldsymbol{X}_i)}\{Y_i - \hat{m}_a(\boldsymbol{X}_i)\} - \hat{\psi}_a^{\mathsf{PI},2} \right)$$

Above, $m_a(\boldsymbol{X})$ and $\hat{g}_a(\boldsymbol{X})$ can be estimated with <u>weighted regressions</u> that add weights $R/\hat{\mathbb{P}}(R = 1|\boldsymbol{Z})$ to the underlying loss function (Rose and van der Laan, 2011)

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
Y_3	Y_3^*	A_3	A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
Y_5	Y_5^*	A_5	A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



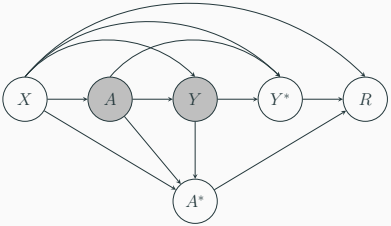
$$\hat{\psi}_a^{\mathsf{OS},2} = \hat{\psi}_a^{\mathsf{PI},2} + \frac{1}{n}\sum_{i=1}^{n}\left(\hat{m}_a(\boldsymbol{X}_i) + \frac{I(A_i = a)}{\hat{g}_a(\boldsymbol{X}_i)}\{Y_i - \hat{m}_a(\boldsymbol{X}_i)\} - \hat{\psi}_a^{\mathsf{PI},2}\right)$$

Issue: The bias correction terms are <u>only observed</u> when $R_i = 1$

- Above estimator is infeasible

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
Y_3	Y_3^*	A_3	A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
Y_5	Y_5^*	A_5	A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



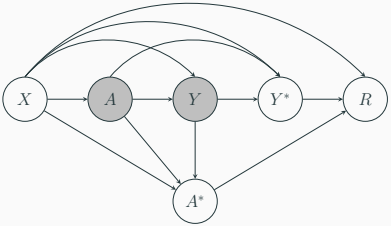
Key idea: Treat the bias correction terms as pseudo-outcomes:

$$\hat{\psi}_a^{\mathsf{OS},2} = \hat{\psi}_a^{\mathsf{PI},2} + \frac{1}{n}\sum_{i=1}^{n}\left(\hat{m}_a(\boldsymbol{X}_i) + \frac{I(A_i = a)}{\hat{g}_a(\boldsymbol{X}_i)}\{Y_i - \hat{m}_a(\boldsymbol{X}_i)\} - \hat{\psi}_a^{\mathsf{PI},2}\right)$$

$$= \hat{\psi}_a^{\mathsf{PI},2} + \frac{1}{n}\sum_{i=1}^{n}\underbrace{\chi_a(\boldsymbol{O}_i; \hat{m}_a, \hat{g}_a)}_{\text{pseudo outcome}}$$

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
Y_3	Y_3^*	A_3	A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
Y_5	Y_5^*	A_5	A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1

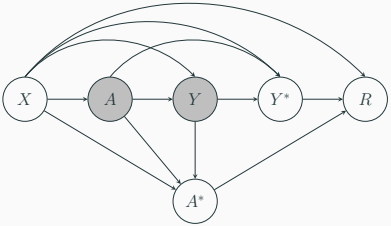


Key idea: Perform AIPW on the pseudo-outcome:

$$\hat{\psi}_a^{\mathsf{OS},2} = \hat{\psi}_a^{\mathsf{PI},2} + \frac{1}{n}\sum_{i=1}^{n} \chi_a(\boldsymbol{O}_i; \hat{m}_a, \hat{g}_a)$$

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
Y_3	Y_3^*	A_3	A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
Y_5	Y_5^*	A_5	A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1

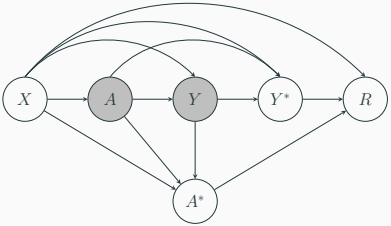


Key idea: Perform AIPW on the pseudo-outcome:

$$\hat{\psi}_a^{\mathsf{OS},2} = \hat{\psi}_a^{\mathsf{PI},2} + \frac{1}{n}\sum_{i=1}^{n}\left(\hat{\varphi}_a(\boldsymbol{Z}_i) + \frac{R_i}{\mathbb{P}(R_i = 1|\boldsymbol{Z}_i)}\{\chi_a(\boldsymbol{O}_i; \hat{m}_a, \hat{g}_a) - \hat{\varphi}_a(\boldsymbol{Z}_i)\}\right)$$

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
Y_3	Y_3^*	A_3	A_3^*	X_3	0
Y_4	Y_4^*	A_4	A_4^*	X_4	1
Y_5	Y_5^*	A_5	A_5^*	X_5	0
Y_6	Y_6^*	A_6	A_6^*	X_6	1



Key idea: Perform AIPW on the pseudo-outcome:

$$\hat{\psi}_a^{\mathsf{OS},2} = \hat{\psi}_a^{\mathsf{PI},2} + \frac{1}{n}\sum_{i=1}^{n}\left(\hat{\varphi}_a(\boldsymbol{Z}_i) + \frac{R_i}{\mathbb{P}(R_i = 1|\boldsymbol{Z}_i)}\{\chi_a(\boldsymbol{O}_i; \hat{m}_a, \hat{g}_a) - \hat{\varphi}_a(\boldsymbol{Z}_i)\}\right)$$

where $\hat{\varphi}_a(\boldsymbol{Z}) = \hat{\mathbb{E}}[\chi_a(\boldsymbol{O}; \hat{m}_a, \hat{g}_a)|\boldsymbol{Z}, R = 1]$

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
Y_2	Y_2^*	A_2	A_2^*	X_2	0
Y_3	Y_3^*	A_3	A_3^*	X_3	0
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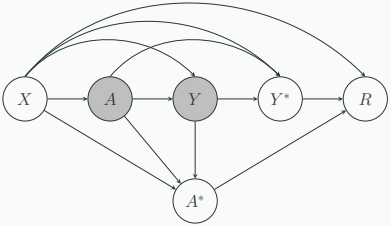

Key idea: Perform AIPW on the pseudo-outcome:

$$\hat{\psi}_a^{\mathsf{OS},2} = \hat{\psi}_a^{\mathsf{PI},2} + \frac{1}{n}\sum_{i=1}^{n}\left(\hat{\varphi}_a(\boldsymbol{Z}_i) + \frac{R_i}{\mathbb{P}(R_i = 1|\boldsymbol{Z}_i)}\{\chi_a(\boldsymbol{O}_i; \hat{m}_a, \hat{g}_a) - \hat{\varphi}_a(\boldsymbol{Z}_i)\}\right)$$

where $\hat{\varphi}_a(\boldsymbol{Z}) = \hat{\mathbb{E}}[\chi_a(\boldsymbol{O}_i; \hat{m}_a, \hat{g}_a)|\boldsymbol{Z}, R = 1]$

- Feasible estimator!
- When sampling probabilities are known, $\hat{\psi}_a^{\mathsf{OS},2}$ is doubly-robust in the traditional sense

Approach 2: Complete-data projection

Y	Y^*	A	A^*	X	R
Y_1	Y_1^*	A_1	A_1^*	X_1	1
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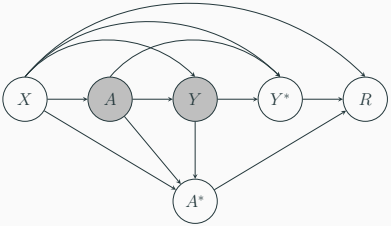

Connections between $\hat{\psi}_a^{\text{OS},1}$ and $\hat{\psi}_a^{\text{OS},2}$:

- Can be shown that asymptotically, Approach 1 and Approach 2 estimators are *equivalent*
  - Approach 2 can be viewed as a re-parameterization of Approach 1
- Both estimators have unique sources of finite sample instability

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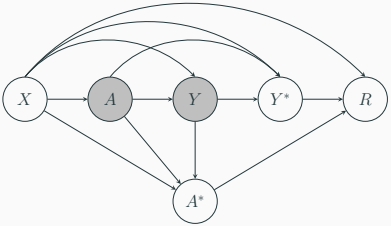

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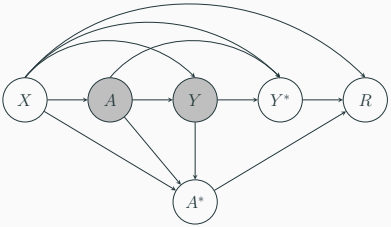

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  - Approach 2: The pseudo-outcome regression function $\varphi_a(\boldsymbol{Z})$ can be difficult to estimate in small samples

Roadmap

Chapter 2

Background

Methods

Approach 1

Approach 2

Ensemble Estimator

Data Application

Discussion

Approaches 1 and 2

Previous slides suggest

- $\hat{\psi}_a^{\mathsf{OS},1}$ and $\hat{\psi}_a^{\mathsf{OS},2}$ have the same asymptotic distribution, but...
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Finite sample performance is crucial in two-phase settings, since phase-two samples will often be small

Challenge: Choosing to report an estimator based on post-hoc efficiency checks invalidates inference

Ensemble Estimator

Can avoid the need to choose an estimator *a priori* by taking a variance-optimal weighted average of the two one-step estimators

$$\hat{\psi}_a^{\mathsf{OS,W}} = \hat{w} \cdot \hat{\psi}_a^{\mathsf{OS,1}} + (1 - \hat{w}) \cdot \hat{\psi}_a^{\mathsf{OS,2}}$$

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Advantage: Provides a means to pre-register an estimator of $\mathbb{E}[Y(a)]$ that is agnostic to whether $\hat{\psi}_a^{\text{OS,1}}$ or $\hat{\psi}_a^{\text{OS,2}}$ will be more efficient

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VCCC Data Application

Recall that in the VCCC EHR database, numerous variables were recorded with error and later validated

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Goal: Compare our proposed Approach 1 and 2 one-step estimators and the ensemble estimator to **(i)** an oracle AIPW estimator which has access to A and Y for all individuals, and **(ii)** a naive AIPW estimator which ignores measurement error

VCCC application

"Reveal" increasingly larger shares of the validated data $\mathbb{P}(R = 1)$, over-sampling those with $A^* = 1$ and $Y^* = 1$

- E.g. 5%, 10%, $15\%, \ldots$
- Implement proposed estimators at each share

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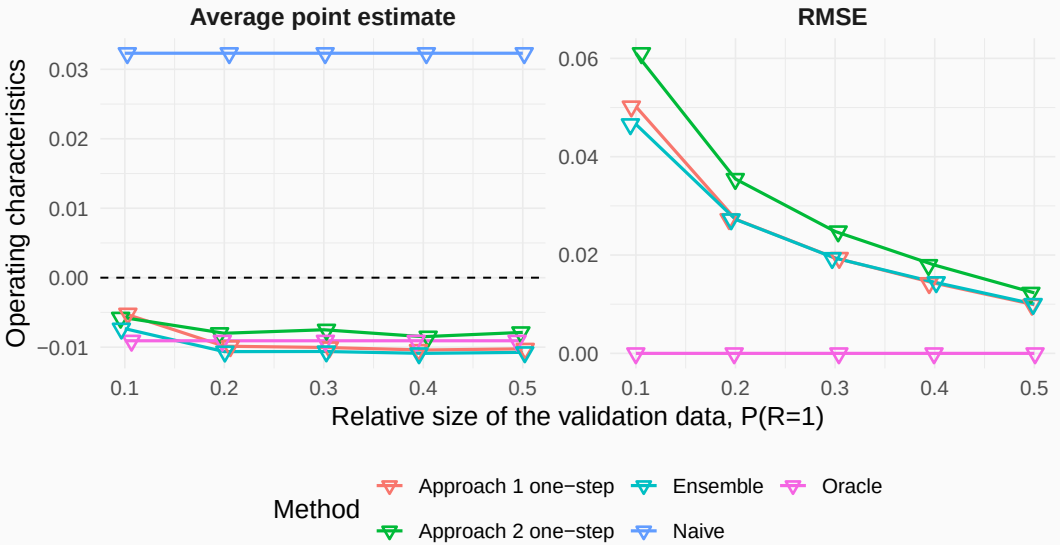
Operating characteristics: 3−year ADE risk

Treatment: ART initiation within one month of first visit



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- Methods
  - Approach 1
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Discussion

Growth in use of biased validation sampling schemes, particularly in the presence of joint outcome and treatment measurement error

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Growth in use of biased validation sampling schemes, particularly in the presence of joint outcome and treatment measurement error

We've presented debiased one-step estimators that accommodate these sampling schemes (and general two-phase sampling schemes)

- Ensemble estimator will tend to be as or more efficient than the better of $\hat{\psi}_a^{\mathsf{OS},1}$ and $\hat{\psi}_a^{\mathsf{OS},2}$ in finite samples
- We developed the R package `drcmd`, which implements Approach 2 for general missing data patterns

Roadmap

Introduction

Chapter 1

Chapter 2

Chapter 3

Discussion

Debiased Machine Learning for Conformal Prediction of Counterfactual Outcomes Under Runtime Confounding

Chapter 3 Go to Appendix

with Kevin P. Josey, Rachel C. Nethery and Giovanni Parmigiani

Under review

Roadmap

Chapter 3

Background

Counterfactual Prediction Intervals

Runtime Confounding and Multi-Source Data

Problem setting

Data Structure

Conformal Prediction Primer

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Simulation

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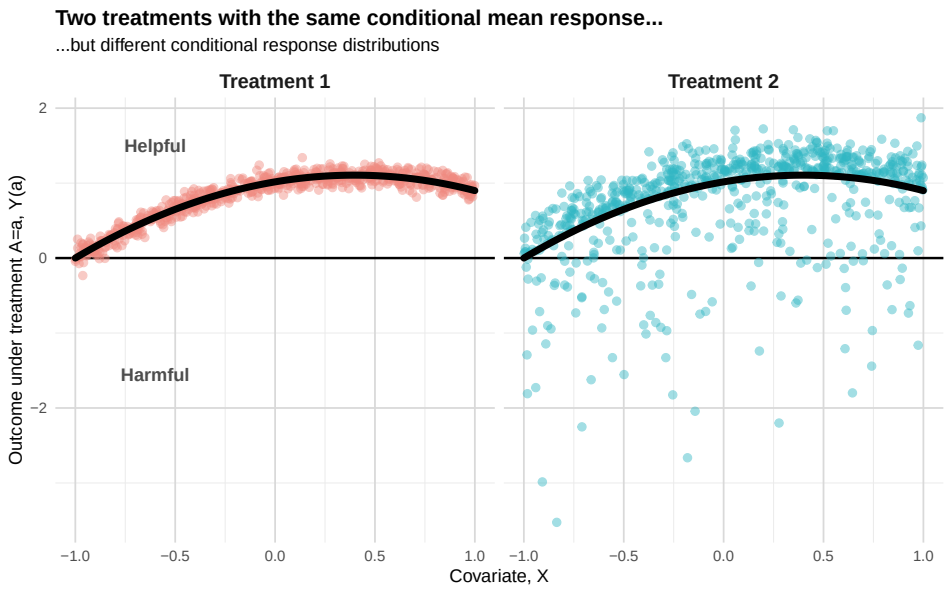
Along with point predictions, prediction intervals can serve as key inputs for decision making

Counterfactual predictions vs. prediction intervals



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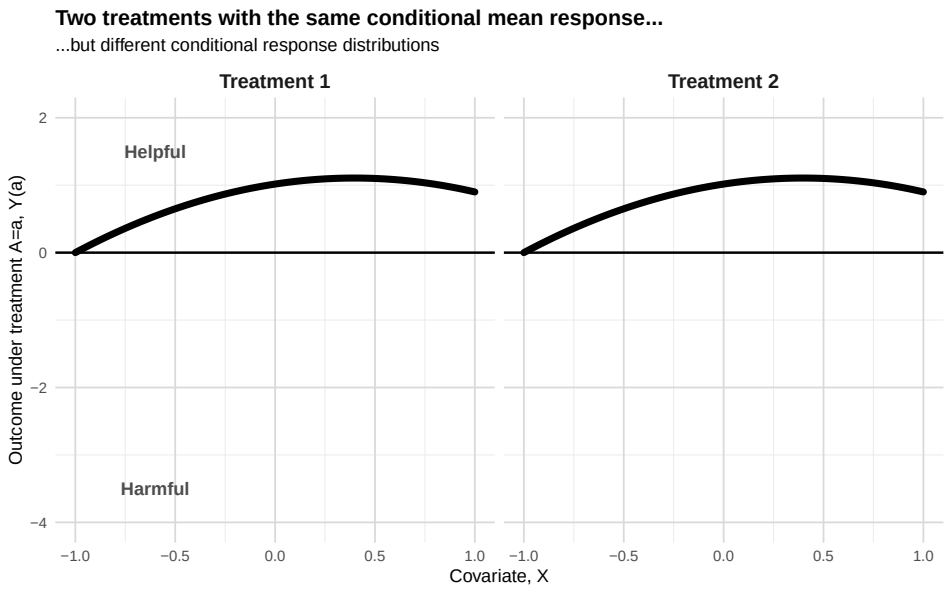

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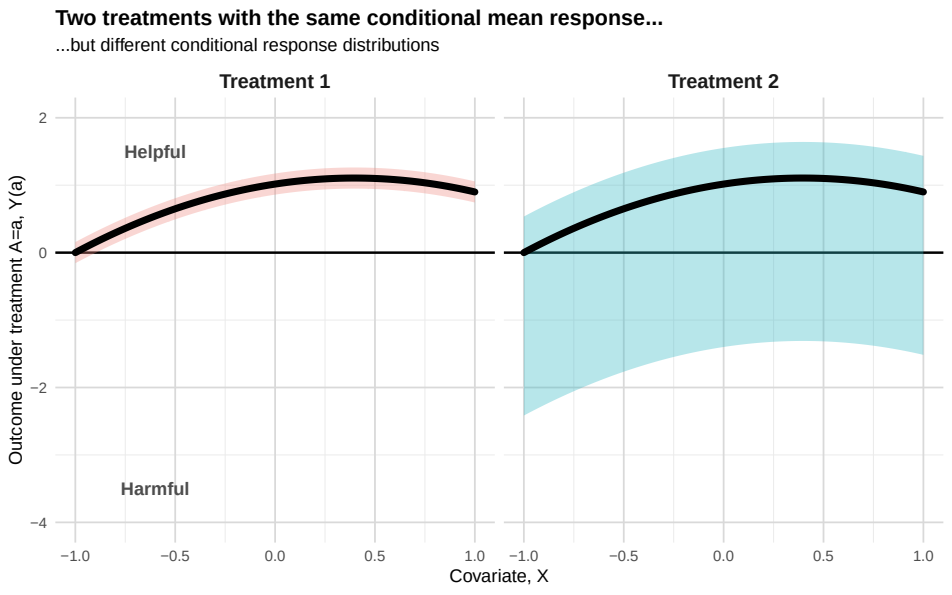

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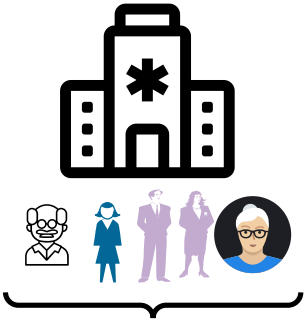
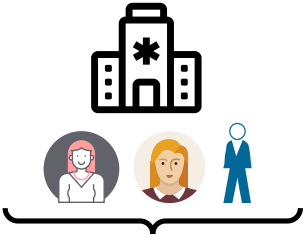


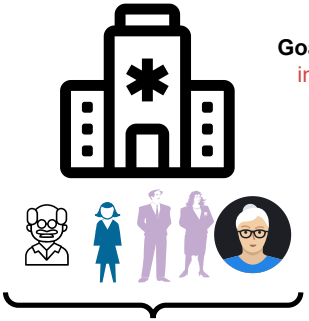
$$(Y_i, A_i, \mathbf{X}_i)_{i=1}^{n_1}$$



$$(\mathbf{V}_i)_{i=1}^{n_0}$$

where $\mathbf{V} \subset \mathbf{X}$

Goal: use info from source site to inform decisions in target site



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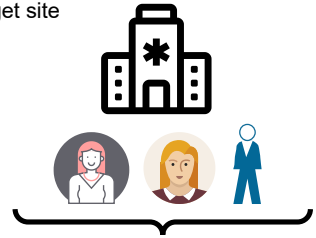

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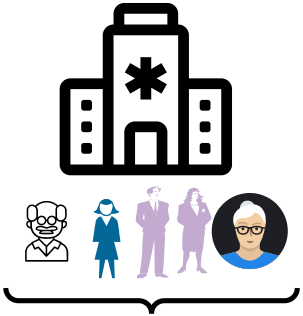

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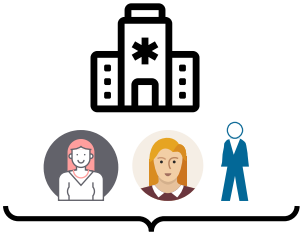

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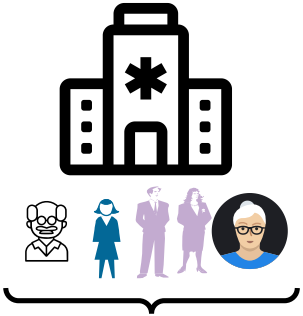

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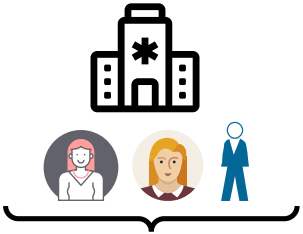

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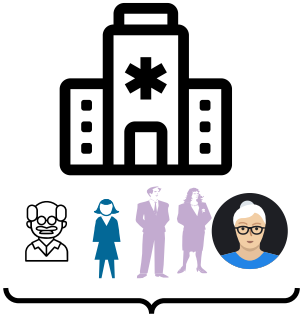

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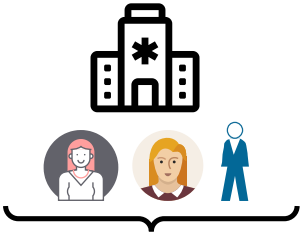

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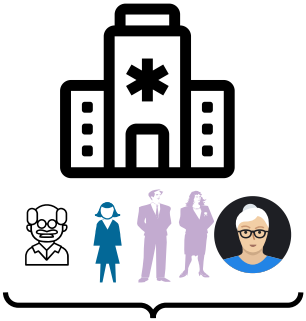

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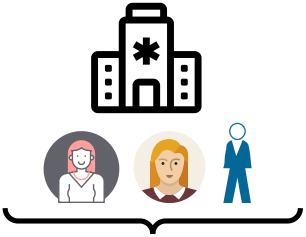

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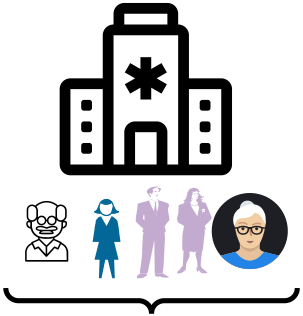

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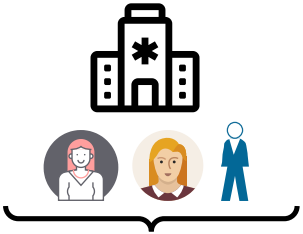

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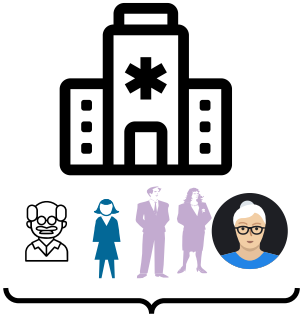
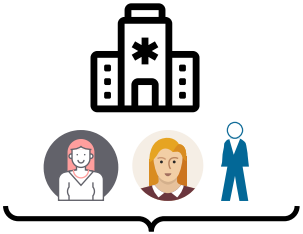


$$(Y_i, A_i, \mathbf{X}_i)_{i=1}^{n_1}$$



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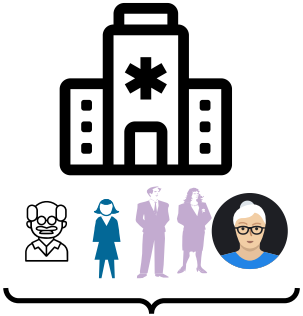
$$\hat{m}_a(\mathbf{V})$$



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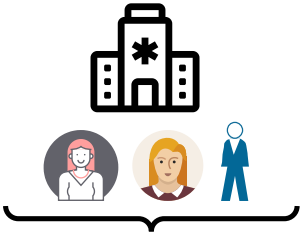

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$$\hat{m}_a(\mathbf{V})$$

$\mathbf{V}$ may not contain all confounders

This scenario is commonly referred to as
runtime confounding (Coston et al. 2020)

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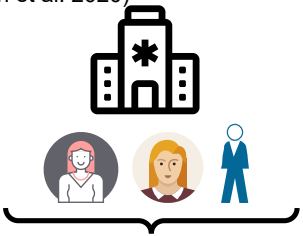

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where $\mathbf{V} \subset \mathbf{X}$

$$\hat{m}_a(\mathbf{V})$$

Contributions

This paper...

- Proposes methods for constructing prediction intervals for counterfactual outcomes $Y(a)$ under runtime confounding

- We construct these intervals in a model-agnostic way that
  - Ensures their validity asymptotically, and
  - Relies on less data to obtain $\approx$ valid coverage than commonly-taken approaches

Roadmap

Chapter 3

Background

Counterfactual Prediction Intervals

Runtime Confounding and Multi-Source Data

Problem setting

Data Structure

Conformal Prediction Primer

Methods

Simulation

Roadmap

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Background

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Problem setting

Data Structure

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Methods

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Suppose we observe...

- $(Y_i, A_i, \boldsymbol{X}_i),\ i = 1, \ldots, n_1$, from a <u>source population</u>
- $\boldsymbol{V}_i,\quad i = 1, \ldots, n_0$ from a <u>target population</u>, where $\boldsymbol{V}$ is a subset of all the covariates collected in $\boldsymbol{X}$, where $\boldsymbol{X} = (\boldsymbol{V}, \boldsymbol{U})$

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Interest lies in constructing prediction intervals for counterfactual outcomes in the target population which attain a desired marginal coverage probability of $1 - \alpha$:

$$\mathbb{P}\{\ Y_i(a) \in \hat{C}_a(\boldsymbol{V}_i) \mid \text{In target pop.}\ \} \geq 1 - \alpha, \quad a \in \mathcal{A}$$

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Y	A	$\boldsymbol{V}$	$\boldsymbol{U}$
Y_1	A_1	$\boldsymbol{V}_1$	$\boldsymbol{U}_1$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
Y_{n_1}	A_{n_1}	$\boldsymbol{V}_{n_1}$	$\boldsymbol{U}_{n_1}$

Source sample

Y	A	$\boldsymbol{V}$	$\boldsymbol{U}$
NA	NA	$\boldsymbol{V}_{n_1+1}$	NA
$\vdots$	$\vdots$	$\vdots$	$\vdots$
NA	NA	$\boldsymbol{V}_{n_1+n_0}$	NA

Target sample

Problem setting

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Y_1	A_1	$\boldsymbol{V}_1$	$\boldsymbol{U}_1$	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
Y_{n_1}	A_{n_1}	$\boldsymbol{V}_{n_1}$	$\boldsymbol{U}_{n_1}$	1
NA	NA	$\boldsymbol{V}_{n_1+1}$	NA	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
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This is an example of runtime confounding: the covariates we have access to at test time differ from those available to us at training time (Coston et al. 2020)

Counterfactual prediction intervals

Y	A	$\boldsymbol{V}$	$\boldsymbol{U}$	S
Y_1	A_1	$\boldsymbol{V}_1$	$\boldsymbol{U}_1$	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
Y_{n_1}	A_{n_1}	$\boldsymbol{V}_{n_1}$	$\boldsymbol{U}_{n_1}$	1
NA	NA	$\boxed{\boldsymbol{V}_{n_1+1}}$	NA	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
NA	NA	$\boldsymbol{V}_{n_1+n_0}$	NA	0

Fit model for Y(a)

$$\widehat{C}_a\left(V_{n_1+1}\right)$$

Interval for Y(a)

Assumptions

1. **(Positivity in the source study)**: $0 < \mathbb{P}(A = a | \boldsymbol{X}, S = 1) < 1$
2. **(Consistency)**: $Y = \sum_{a \in \mathcal{A}} I(A = a) \cdot Y(a)$
3. **(Positivity of source membership)**: $0 < \mathbb{P}(S = 1 | \boldsymbol{V}) < 1$
4. **Runtime Confounding in Source**: $Y(a) \perp\!\!\!\perp A | \boldsymbol{X}, S = 1$, but $Y(a) \not\perp\!\!\!\perp A | \boldsymbol{V}, S = 1$
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Roadmap

Chapter 3

Background

Counterfactual Prediction Intervals

Runtime Confounding and Multi-Source Data

Problem setting

Data Structure

Conformal Prediction Primer

Methods

Simulation

Conformal prediction: a tool for constructing valid prediction intervals

Conformal prediction (Vovk et al., 2005) is a general framework that

- Takes an arbitrary prediction function $\hat{f}(\boldsymbol{X})$ for an outcome Y, and
- Spits out a prediction interval function $\hat{C}(\boldsymbol{X})$ for new observations such that

$$\mathbb{P}(Y \in \hat{C}(\boldsymbol{X})) \geq 1 - \alpha$$

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When training + testing data are from the same distribution, above guarantee holds in finite samples (!)

In more general settings, this guarantee only holds asymptotically

- Large set of existing work focuses on methods for "speeding up" the asymptotics

The key ingredient in conformal prediction

The key idea behind conformal prediction (no covariate shift, everything i.i.d.) is to

1. Obtain absolute residuals $R(\boldsymbol{X}_i, Y_i) = |Y_i - \hat{f}(\boldsymbol{X}_i)|$ (often called conformity scores)
2. Let $\hat{C}(\boldsymbol{X}_i) = \hat{f}(\boldsymbol{X}_i) \pm r_\alpha$, where r_α is the $1-\alpha$ quantile of the abs. residuals
3. Notice this will imply for a new observation that $\mathbb{P}(Y \in \hat{C}(\boldsymbol{X})) \geq 1 - \alpha$

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This is because

$$\{Y \in \hat{C}(\boldsymbol{X})\} = \{\text{The residual } R_i \leq r_\alpha\} = \{\text{The residual } R_i \text{ falls below its } 1-\alpha \text{ quantile}\}$$

Notice the probability of the last event is $1 - \alpha$

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For our problem of interest, this means we seek an interval $\hat{C}_a(\boldsymbol{V}) = \hat{\mu}_a(\boldsymbol{V}) \pm r_{a,\alpha}$ such that

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Key problem: <u>We don't observe outcomes</u>, treatments, or all of $\boldsymbol{X}$ in the target pop, so we can't directly estimate $r_{a,\alpha}$ for our target population

- No way to compute residuals R_i, so there's no way to directly estimate $r_{a,\alpha}$

Roadmap

Chapter 3

Background

Counterfactual Prediction Intervals

Runtime Confounding and Multi-Source Data

Problem setting

Data Structure

Conformal Prediction Primer

Methods

Simulation

Quick aside: forming counterfactual predictions

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Coston et al. (2020) provide useful framework

1. First, fit $\hat{m}_a(\boldsymbol{X}) = \hat{\mathbb{E}}(Y|A = a, \boldsymbol{X})$ with the training data
2. Next, regress the estimated $\hat{m}_a(\boldsymbol{X})$ on $\boldsymbol{V}$, yielding $\hat{\mu}_a(\boldsymbol{V}) = \hat{\mathbb{E}}(\hat{m}_a(\boldsymbol{X})|\boldsymbol{V})$

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Approach is motivated by the observation that

$$\begin{aligned}\mathbb{E}(Y(a)|\boldsymbol{V}) \;=\; \mathbb{E}[\,\mathbb{E}(Y(a)|\underbrace{\boldsymbol{V},\boldsymbol{U}}_{\boldsymbol{X}})\,|\boldsymbol{V}] \;&=\; \mathbb{E}(\;\mathbb{E}(Y|A=a,\boldsymbol{X})\;\;|\boldsymbol{V}) \\ &=\; \mathbb{E}(m_a(\boldsymbol{X})\mid \boldsymbol{V})\end{aligned}$$

Interval construction relies on estimation of $r_{a,\alpha}$

Suppose we've fit a prediction model $\hat{\mu}_a(\boldsymbol{V})$ for $\hat{Y}(a)$, let $R_a = |Y(a) - \hat{\mu}_a(\boldsymbol{V})|$, and recall our interest in the equation

$$\mathbb{P}(R_a(\boldsymbol{V}_i, Y_i) \leq r_{a,\alpha} | S = 0) = 1 - \alpha$$

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- Our paper focuses on estimation of $r_{a,\alpha}$, with particular attention given to semiparametric efficient estimation

IPW estimator of $r_{a,\alpha}$

Continuing to let $r_{a,\alpha}$ be the quantity which satisfies

$$\mathbb{P}(R(\boldsymbol{V}_i, Y_i(a)) \leq r_{a,\alpha} | S = 0) = 1 - \alpha$$

It can be shown that this same quantity *also* satisfies

$$\mathbb{E}\left[\underbrace{\frac{\mathbb{P}(S = 0|\boldsymbol{V})}{\mathbb{P}(A = a|\boldsymbol{X}, S = 1)\mathbb{P}(S = 1|\boldsymbol{V})}}_{\text{weights}} \underbrace{SI(A = a)\mathbb{I}\{R_a(\boldsymbol{V}, Y(a)) \leq r_{a,\alpha}\}}_{\text{source pop. conformity scores}} \right] = 1 - \alpha$$

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Intuition: (1) Take the scores that you compute among those with $A = a$ in the source pop, and (2) find the $1 - \alpha$ quantile of its reweighted distribution

Stepping back

- Remember that our goal is to construct intervals for $Y(a)$ in the target population so that

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- So our goal (bullet 1) boils down to estimating $r_{a,\alpha}$ well

Slow convergence of the plug-in estimator

The earlier weighting expression suggests we can estimate $r_{a,\alpha}$ via the estimating equation

$$\frac{1}{n}\sum_{i=1}^{n}\left(\frac{\hat{\mathbb{P}}(S_i=0|\boldsymbol{V}_i)}{\hat{\mathbb{P}}(A_i=a|\boldsymbol{X}_i,S_i=1)\hat{\mathbb{P}}(S_i=1|\boldsymbol{V}_i)}\underbrace{S_i I(A_i=a)\mathbb{I}\{R_a(Y_i,\boldsymbol{V}_i)\leq r_{a,\alpha}\}}_{\text{source pop scores}}\right)-(1-\alpha)=0$$

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But this suggests the rate at which we estimate $r_{a,\alpha}$ will be <u>dictated</u> by

$$\hat{r}_{a,\alpha}-r_{a,\alpha}=O_{\mathbb{P}}(1/\sqrt{n}+||\hat{g}_a-g_a||+||\hat{\kappa}-\kappa||)$$

where $\hat{g}_a(\boldsymbol{X})=\hat{\mathbb{P}}(A=a|S=1,\boldsymbol{X})$ and $\hat{\kappa}(\boldsymbol{V})=\hat{\mathbb{P}}(S=1|\boldsymbol{V})$

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If we use flexible models, this will be much slower than ideal $\sqrt{n}$ rates

- We've made the problem of estimating $r_{a,\alpha}$ as hard as the problem of estimating the above nuisance functions $\rightarrow$ shouldn't it be easier?

Debiased estimator

Much like in ATE estimation, we can make the problem of estimating $r_{a,\alpha}$ easier by basing estimation on its EIF. Letting η collect all nuisance functions, we can choose $\hat{r}_{a,\alpha}$ so

$$\frac{1}{n}\sum_{i=1}^{n} \mathsf{EIF}(\hat{r}_{a,\alpha},\ \mathsf{data}_i,\ \hat{\eta}) = 0$$

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Theorem If all the nuisance functions making up η are estimated at $n^{-1/4}$ rates,

$$\mathbb{P}(Y(a) \in \hat{C}_a(\boldsymbol{V}; \hat{r}_{a,\alpha}) | S = 0) = 1 - \alpha + O_{\mathbb{P}}(1/\sqrt{n})$$

Debiased estimator

Much like in ATE estimation, we can make the problem of estimating $r_{a,\alpha}$ easier by basing estimation on its EIF. Letting η collect all nuisance functions, we can choose $\hat{r}_{a,\alpha}$ so

$$\frac{1}{n}\sum_{i=1}^{n} \text{EIF}(\hat{r}_{a,\alpha},\ \text{data}_i,\ \hat{\eta}) = 0$$

Why? The population-level EIF for the true $r_{a,\alpha}$ (and any statistical functional with an EIF) is mean zero

Theorem If all the nuisance functions making up η are estimated at $n^{-1/4}$ rates,

$$\mathbb{P}(Y(a) \in \hat{C}_a(\boldsymbol{V}; \hat{r}_{a,\alpha}) | S = 0) = 1 - \alpha + O_{\mathbb{P}}(1/\sqrt{n})$$

Main idea: EIF-based estimation dramatically improves how "fast" we can estimate $\hat{r}_{a,\alpha}$, which in turn speeds up coverage properties

Brief snapshot of simulation study

Compared our proposed method to

- Weighted conformal prediction (Tibshirani et al., 2019) based on earlier IPW expression
- A doubly-robust method based on Yang et al. (2024) that *ignores* runtime confounding, pretending $\boldsymbol{V}$ collects all confounders

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Simulated data corresponding to a runtime confounding scenario, varying the sample size

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Simulated data corresponding to a runtime confounding scenario, varying the sample size

For simplicity, focused on producing 90% prediction intervals for $Y(1)$ in target data. Across 1,000 iterations for each sample size considered, we compared

- The average coverage rate of $Y(1)$ in target data
- The average interval width

Roadmap

Chapter 3

Background

Counterfactual Prediction Intervals

Runtime Confounding and Multi-Source Data

Problem setting

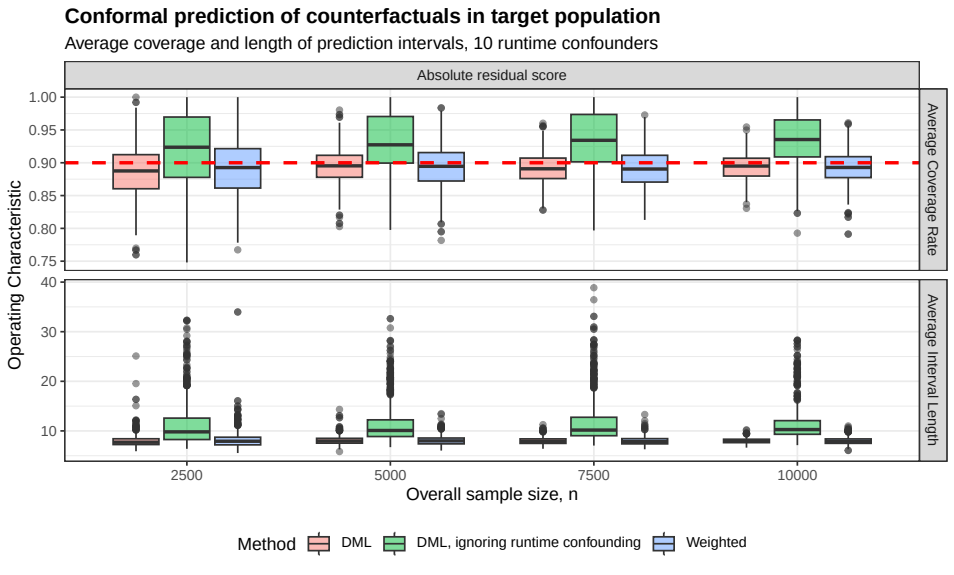
Data Structure

Conformal Prediction Primer

Methods

Simulation

Brief snapshot of simulation study



Roadmap

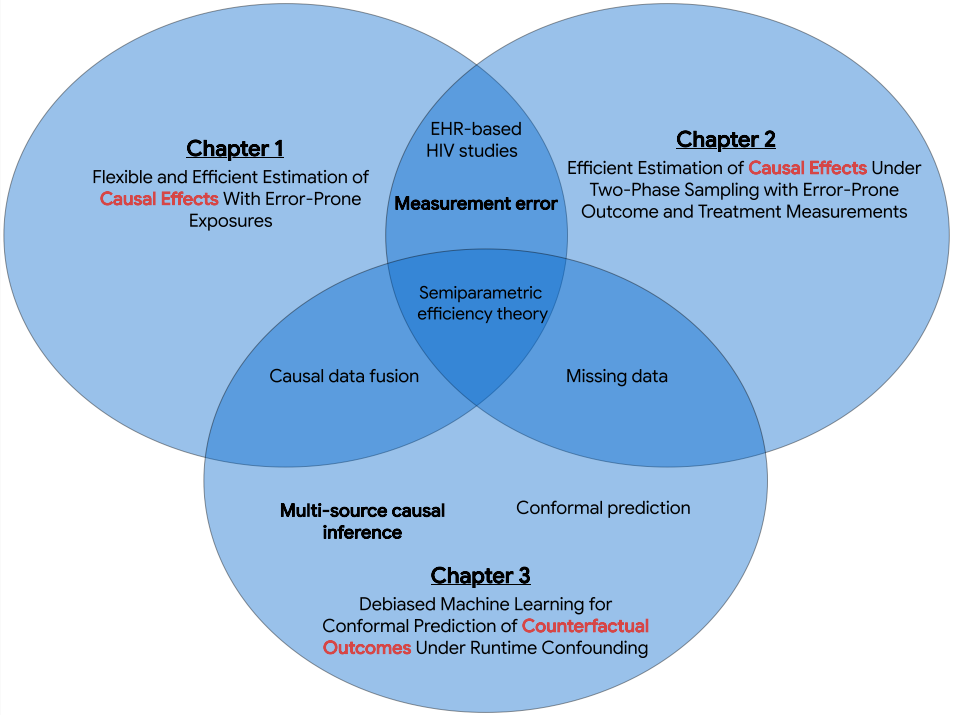
Introduction

Chapter 1

Chapter 2

Chapter 3

Discussion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EHR-based HIV studies
Measurement error
Chapter 2
Efficient Estimation of Causal Effects Under Two-Phase Sampling with Error-Prone Outcome and Treatment Measurements
Semiparametric efficiency theory
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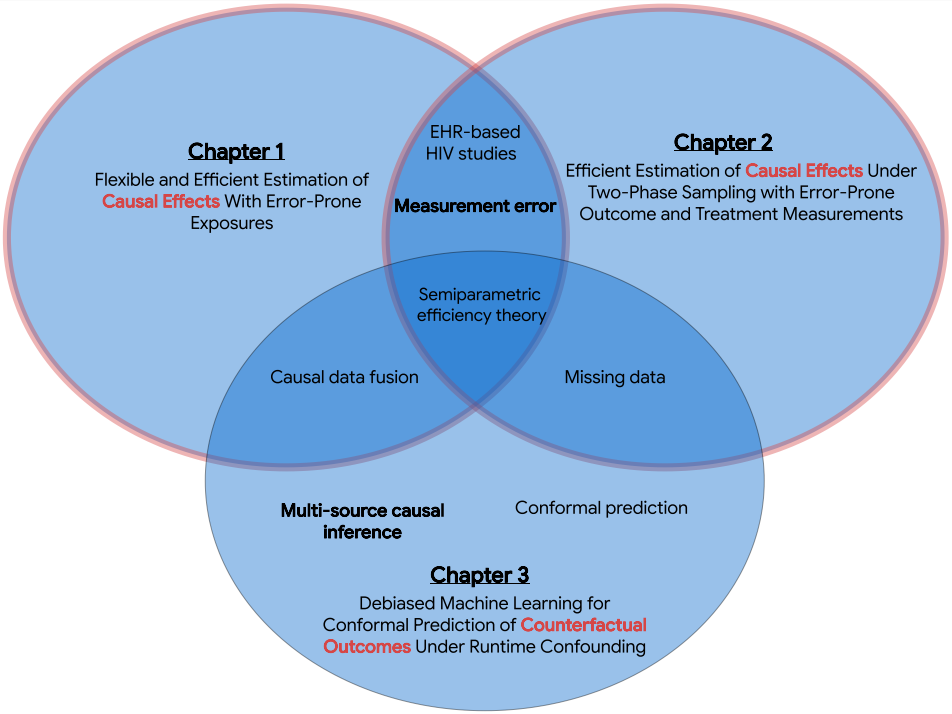
Debiased Machine Learning for Conformal Prediction of Counterfactual Outcomes Under Runtime Confounding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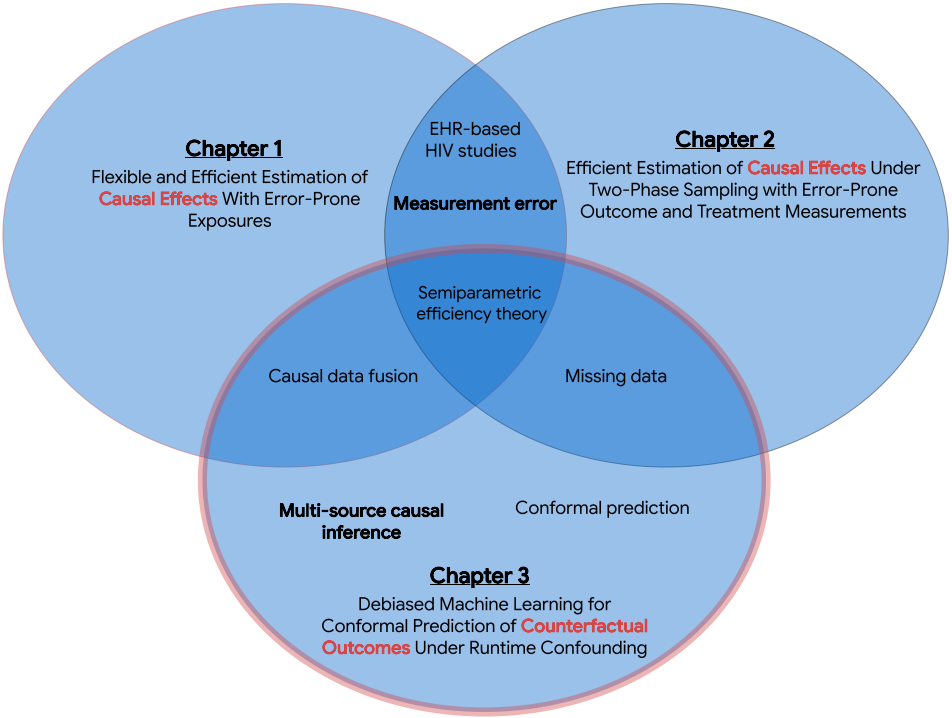
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Thank you!

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